



MOTION DETECTION SYSTEM USING PIR SENSOR AND ARDUINO

Task 3: Mini Project

Submitted By

Name: Chaitanya Reddy Paturu

Internship Domain: Internet of Things (IoT)

Organization: CodeAlpha



[DATE]

[COMPANY NAME]

[Company address]

Motion Detection System Using PIR Sensor and Arduino

Abstract

Motion detection systems are widely used in security applications, smart homes, automatic lighting systems, and surveillance systems. This project uses a Passive Infrared (PIR) Sensor and Arduino Uno to detect human movement. When motion is detected, the PIR sensor sends a signal to the Arduino, which activates an LED. This project demonstrates a simple and cost-effective motion detection system that can be extended to various real-world applications.

1. Introduction

Security and automation have become important aspects of modern technology. Motion detection systems help monitor environments and detect the presence of humans or moving objects. A Passive Infrared (PIR) sensor detects infrared radiation emitted by living beings. By interfacing the PIR sensor with an Arduino Uno, motion can be detected and an output device such as an LED can be controlled automatically. This project demonstrates the implementation of a basic motion detection system using IoT concepts and embedded systems.

2. Objective

- To detect human motion using a PIR sensor.
 - To interface the PIR sensor with Arduino Uno.
 - To control an LED based on motion detection.
 - To understand sensor-based automation systems.
-

3. Components Required

S.No	Component
1	Arduino Uno
2	PIR Motion Sensor
3	LED
4	220Ω Resistor
5	Breadboard
6	Jumper Wires
7	USB Cable

4. Theory

The PIR (Passive Infrared) sensor detects changes in infrared radiation within its sensing range.

Human bodies emit infrared radiation naturally. When a person moves in front of the PIR sensor, the sensor detects the change and produces a HIGH output signal.

The Arduino continuously monitors the PIR output pin. When motion is detected, the Arduino turns ON the LED. When no motion is detected, the LED remains OFF.

5. Circuit Diagram

PIR Sensor

VCC -----> 5V
GND -----> GND
OUT -----> D2

Arduino D13 ----220Ω---->| LED ---- GND

6. Working Principle

1. PIR sensor continuously monitors surroundings.
 2. Sensor detects infrared radiation changes.
 3. Output pin becomes HIGH when motion is detected.
 4. Arduino reads the signal from the PIR sensor.
 5. Arduino turns ON the LED.
 6. When motion stops, output becomes LOW.
 7. Arduino turns OFF the LED.
-

7. Arduino Program

```
int pirPin = 2;
int ledPin = 13;

void setup() {
  pinMode(pirPin, INPUT);
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {

  int motion = digitalRead(pirPin);

  if(motion == HIGH) {
    digitalWrite(ledPin, HIGH);
```

```
    Serial.println("Motion Detected");
}
else {
    digitalWrite(ledPin, LOW);
    Serial.println("No Motion");
}

delay(500);
}
```

8. Output

Case 1: No Motion Detected

- PIR Output = LOW
- LED = OFF
- Serial Monitor displays "No Motion"

Case 2: Motion Detected

- PIR Output = HIGH
 - LED = ON
 - Serial Monitor displays "Motion Detected"
-

9. Applications

- Home Security Systems
 - Smart Lighting Systems
 - Intruder Detection
 - Automatic Door Systems
 - Office Security
 - Surveillance Systems
 - Smart Buildings
-

10. Advantages

- Low Cost
 - Easy to Implement
 - Low Power Consumption
 - Reliable Operation
 - Automatic Detection
 - Suitable for Smart Homes
-

11. Limitations

- Limited Detection Range
 - Sensitive to Environmental Conditions
 - Cannot Identify Specific Objects
 - May Require Calibration
-

12. Future Scope

Future enhancements can include:

- IoT Cloud Monitoring
 - Mobile Notifications
 - GSM Alert System
 - Smart Camera Integration
 - Smart Home Automation
 - AI-based Motion Analysis
-

13. Conclusion

The Motion Detection System using PIR Sensor and Arduino successfully detects movement and activates an LED. The project demonstrates the use of sensors, microcontrollers, and automation techniques. It provides a foundation for developing advanced security and smart home applications.

References

1. Arduino Official Documentation
2. PIR Sensor Datasheet
3. Tinkercad Circuit Simulations
4. IEEE Research Papers on Motion Detection Systems
5. Embedded Systems and IoT Learning Resources